7 Read the following article and then answer the questions that follow.

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Powering the world's energy needs through nuclear fission

Nuclear power plants are a type of power plant that use the process of nuclear fission in order to generate electricity. They do this by using nuclear reactors in combination with the Rankine cycle, where the heat generated by the reactor converts water into steam, which spins a turbine and a generator.

The large amount of energy released in a nuclear fission reaction, together with the emission of more than one neutron, has made it possible for neutron-induced fission to be used as a source of useful energy. For most nuclear reactor, the uranium fuel is assembled in such a way that a controlled fission chain reaction can be achieved. When a neutron is captured by a Uranium-235 nucleus, it causes the nucleus to fission. On average, 2.5 neutrons are emitted in these fission reactions. When the conditions are suitable, a chain reaction can occur as shown in Fig. 7.1. If the chain reaction is controlled in the reactor, a source of continuous power may be created.

The induced fission reaction of Uranium-235 may be represented by a nuclear equation of the form

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{a}_{b}\text{P} + ^{c}_{d}\text{Q} + (2 \text{ or } 3)\,^{1}_{0}\text{n} + \text{energy}$$

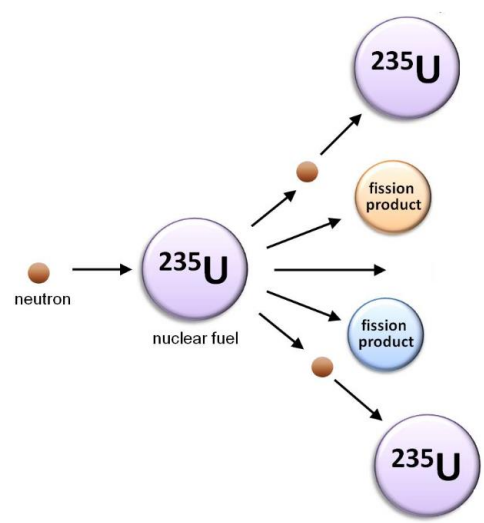

Fig. 7.1

Usually when any two nuclei are fissioned, the fission products may not be the same. If a large sample of Uranium-235 is fissioned, many different fission products will be produced. The percentage amount of each fission product in the fissioned material is referred to as percentage yield.

In 2016, nuclear plants have managed to supply 2477 TWh of electricity worldwide. That was the fourth consecutive year that the global nuclear generation has risen. Nuclear energy generated via nuclear reactions generate heat to produce steam, which is used to generate electricity. The efficiency of a nuclear power plant is determined similarly to other heat engines, since technically the plant is a large heat engine. The amount of electric power produced for each unit of thermal power gives the plant its thermal efficiency. Typical nuclear power plants achieve efficiency of 35%. In addition, nuclear power plants had an average capacity factor of 92.3 percent which is relatively much higher than other electrical systems, meaning they operated at full power on 336 out of 365 days per year.

(a) Suggest why there is a need for the fission process in the nuclear reactor to be controlled.

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…… [2]

(b) Fig. 7.2 shows the variation of the energy release *E* during each stage of the chain reaction with its stage number *n*.

(Stage *n* = 1 represents the stage where the energy is released when the first neutron interacts with first Uranium nucleus and releases 2 or 3 neutrons.
Stages *n* = 2,3… represent the stages where the energy is released when the neutrons from previous stage interacts with other Uranium nuclei and releases 2 or 3 more neutrons for each reaction)

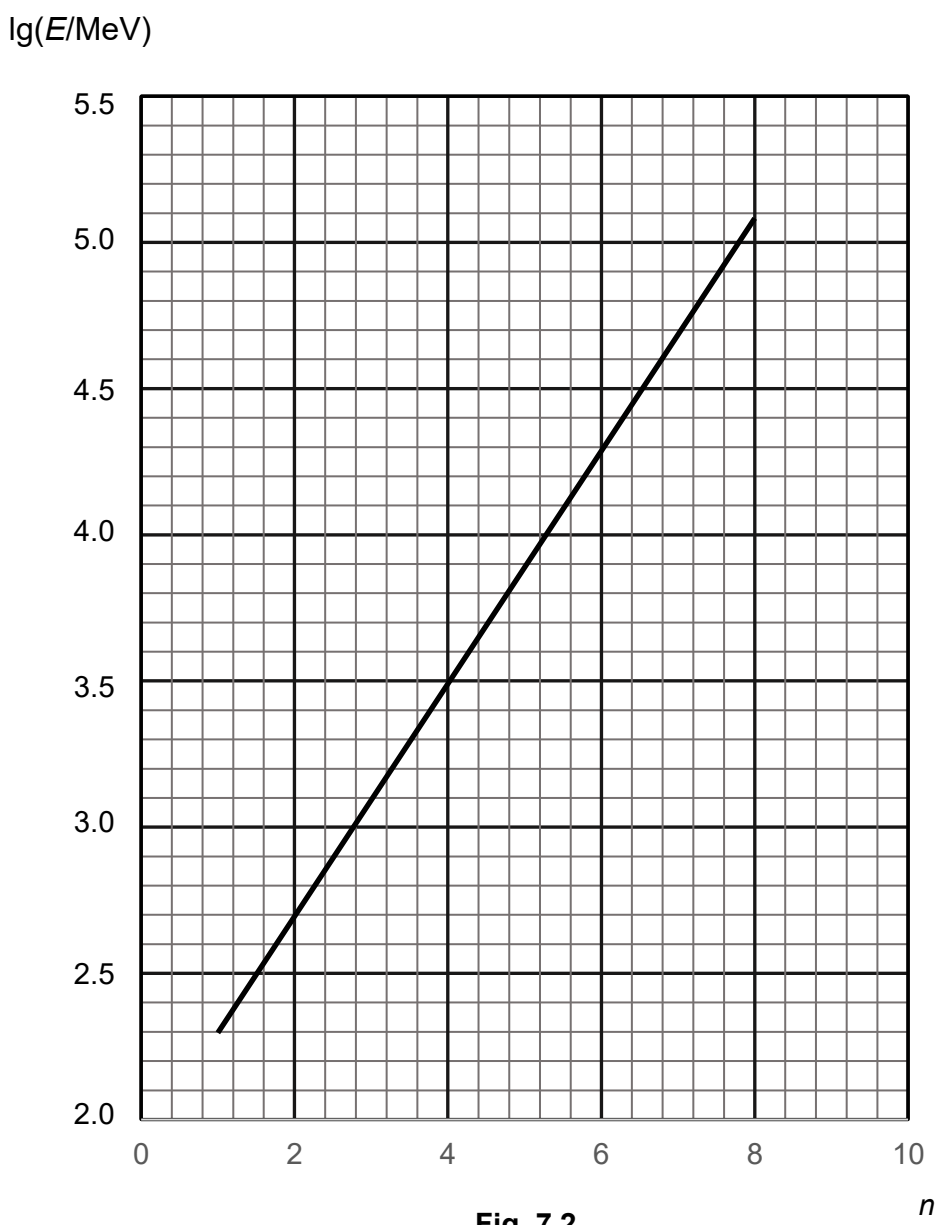

Fig. 7.2

(i) The total energy released per chain E would thus increase with the number n of stage that the fission has proceed till and is given by

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$$E = a^{n-1}b$$

where a is the average number of neutrons emitted per fission reaction and b is a constant.

Use Fig. 7.2 to determine the value of b.
Show your working clearly.

b = …… MeV [3]

(ii) Suggest what is represented by the value of b.

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…… [1]

(c) The variation with nucleon number of the percentage yield of different products is illustrated in Fig. 7.3.

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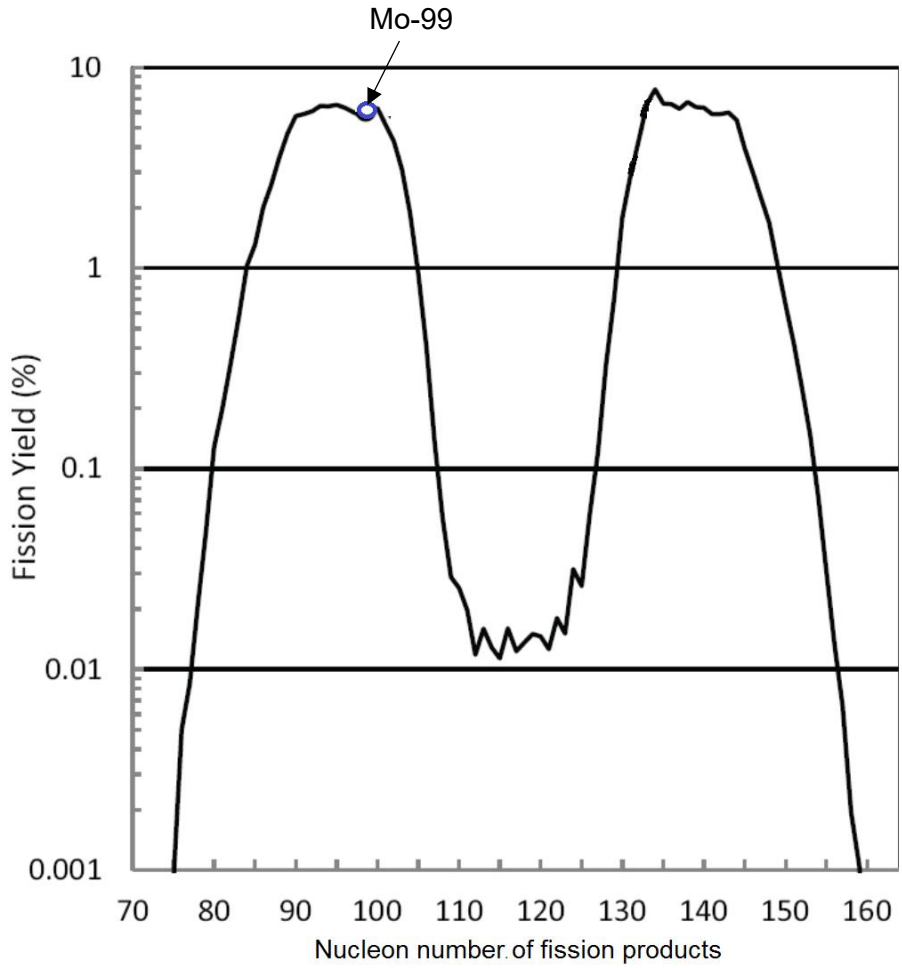

Fig. 7.3

(i) State the proton number of the other fission product for the fission of one uranium if one of the products is $^{89}_{36}Kr$.

proton number = ... [1]

(ii) Suggest why the percentage yield is shown on a logarithmic scale.

.....

..... [1]

(iii) Show that the percentage yield of Mo-99 is about 400 times more than those having masses equal to each other.

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[2]

(d) **(i)** The energy released during one fission reaction of a uranium nuclei occurs partly as kinetic energy of the fission products (167 MeV) and of the neutrons.

Suggest one other mechanism by which energy is released in the fission reaction.

……..…….. [1]

(ii) Determine the amount of heat required to generate the electrical energy worldwide in 2016.

energy = …….. J [2]

(iii) Determine the rate at which uranium is consumed to generate the amount of heat required in 2016.

amount required = …….. kg s^{-1} [3]

(e) The fission products are usually radioactive and give rise to a series of radioactive decay products. Each decay product has its own half-life. Two such fission products with their decay products and half-lives are shown below.

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$$^{90}_{42}\text{Mo} \xrightarrow{67\text{ hours}} {}^{99}_{43}\text{Tc} \xrightarrow{2 \times 10^5\text{ years}} {}^{90}_{44}\text{Ru (stable solid)}$$

$$^{140}_{54}\text{Xe} \xrightarrow{16\text{ seconds}} {}^{140}_{55}\text{Cs} \xrightarrow{1.1\text{ minutes}} {}^{140}_{56}\text{Ba} \xrightarrow{13\text{ days}} {}^{140}_{57}\text{La} \xrightarrow{40\text{ hours}} {}^{140}_{58}\text{Ce (stable solid)}$$

Consider equal amounts of these two products.
Suggest why there are very different problems for the storage of this nuclear waste.

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(f) Despite the advantages of nuclear power, it is unlikely that our world will be purely powered by nuclear energy.
Suggest two reasons, aside from the danger from nuclear waste and cost, for this.

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End of Paper 2